Actl4001 2026 Term 1 Assignment

Lisa Cheng z5482337
Yi Yao Wang z5478775
Connie Gu z5478009
Deetya Jugnarain z5480176

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1 Executive Summary

In line with project objectives, our actuarial assessment aims to recommend insurance products that provide financially sustainable protection across Cosmic Quarry Mining Corporation's (CQMC) four hazard areas while reflecting expected costs, tail risk, system-specific exposures, and future scalability.

Within this framework, the Galaxy General Equipment Failure Shield (EFS) emerges as a suitable inclusion in Galaxy General Insurance Company's portfolio due to its low-frequency but high-severity risk profile. EFS is an indemnity-based product covering sudden and unforeseen mechanical or operational breakdown of mining equipment. As CQMC operations across multiple solar systems, increased reliance on specialised machinery and harsher operating environments makes structured risk transfer increasingly important. The product design incorporates deductibles, policy limits, and operational exclusions to reduce attritional claims, control extreme tail exposure, and support pricing stability, while remaining flexible enough to reflect differing risk conditions across the solar systems.

Risk associated with cargo loss is characterised by low -frequency but high -severity events, driven primarily by environmental hazards. We recommend the inclusion of the Interstellar Cargo Protection Policy (ICPP), a named perils insurance product that protects high-value mineral shipments transported between mining systems. Since CQMC is expanding its mining operations and interstellar transportation networks, more cargo shipments are exposed to hazards, requiring the need of ICPP to protect policyholders form catastrophic losses.

Business interruptions (BI) indicated a low frequency but high severity profile with most individual claims experiencing no claims and the presence of large individual claim size contributes to the right-skewness meaning extreme loss scenarios remain significant and must be considered during pricing and capital plannings. The proposed insurance is Operational Continuity Cover which covers financial losses from operational downtime such as lost production revenue and ongoing expenses. A time deductible would be included to ensure only significant interruptions are covered whilst a monetary deductible and policy limit would help manage smaller claims and control exposure to large losses.

The risk associated with Workers' Compensation is characterised by relatively high-frequency, moderate-severity claims driven by operational workplace incidents. We recommend the StellarCare Protection Plan (SCPP), which provides indemnity cover for employee injuries arising from mining operations. As CQMC expands its workforce across solar systems, exposure to workplace incidents increases, necessitating structured risk transfer. SCPP incorporates a deductible, waiting period, policy limits and cost-sharing to reduce

attritional claims while maintaining protection against more severe injuries. We adjust premiums by solar system to reflect underlying risk differences, ensuring pricing adequacy and supporting ERM through reduced volatility in workforce-related losses.

2 Product Design

2.1 Galaxy General Equipment Failure Shield (EFS)

The EFS provides indemnity coverage for sudden and unforeseen mechanical or operational breakdown of mining equipment, with claims based on repair or replacement costs. The risk profile is characterised by low frequency but high severity, as 94.1% units experience no failures while a small number generate large losses *(Appendix A.1).* A fixed deductible of Đ 50,000 is applied to remove minor attritional claims and reduce administrative costs, while retaining protection against material losses. This deductible is aligned with the lower quartile of observed claims ensuring that small, frequent losses are absorbed by the insured while the insurer covers financially significant events *(Appendix A.2)*. A per-occurrence policy limit of Đ 400,000 is imposed to cap the insurer's exposure to extreme tail losses, this is set close to the 99.9th percentile claim severity to preserves coverage for most claims while truncating only the most extreme outcomes *(Appendix A.3).* This improves capital protection and pricing stability without materially reducing product usefulness to the insured. Coverage is subject to exclusions for gradual wear and tear, excessive usage beyond operational limits, and inadequate maintenance, ensuring that only sudden and insurable risks are covered.

2.2 Interstellar Cargo Protection Policy (ICPP)

The ICPP only insures cargo shipments damaged or destructed due to high debris density on-route or extreme solar radiation exposure. Policyholders are charged an annual premium that is adjusted depending on route risk rating starting from Đ 5.1 M per exposure period *(Appendix A.4)*. Transport routes with higher hazard ratings are associated with greater cargo loss probabilities, leading to greater premiums charged to protect the insurer. Coverage is provided up to the declared cargo value, subject to a policy limit of Đ 400 M. This cap exceeds the 99th percentile losses of Đ 72 M while limiting exposure to extreme tail risk of Đ 678+ M, ensuring an appropriate balance between coverage adequacy and insurer capital protection. Additionally, a staggered deductible rate is offered, ranging from 5% - 10% for risk levels 1 – 5 to reflect the proportional change in risk. Since riskier routes increase both the likelihood and severity of claim, a larger deductible ensures fair cost-sharing, promotes responsible behaviour, and maintains pricing sustainability.

2.3 Operational Continuity Cover (OCC)

OCC is designed as a partial indemnity cover to provide CQMC protection with sustainable insurer exposures. *Appendix A.5* shows simulated statistics from historical data revealing an average gross claim of Đ765,918 while the average paid claim is approximately Đ146,105, revealing 56.6% of losses remains payable after applying waiting period, deductible, coinsurance and policy limits. The results reveal that the policy still responds to interruption events but filters out smaller and less severe losses. Despite the post-product expected loss remains substantial across the solar systems, the cover provides real protection when interruptions are significant hence, tightening the design reduces tail risk. The 99th percentile loss reduces from Đ8.73 B to approximately Đ1.97 B while the premium declines from Đ11.04 B to Đ2.44 B, allowing the cover to be more affordable and insurable (*Appendix A.6)*.

2.4 StellarCare Protection Plan (SCPP)

The SCPP provides indemnity cover for work-related injuries, compensating lost wages, medical costs, and rehabilitation expenses, subject to policy limits and conditions (QBE Australia, 2024). Historical data indicates high-frequency, short-duration claims (median ≈ 7 days) and a heavy-tailed severity distribution *(Appendix A.7)* and thus targets low value claims while limiting exposure to large losses through a Đ5,000 deductible, 21-day waiting period and Đ30,000 policy limit. Cost-sharing (75% medical and 65% wage replacement) mitigates moral hazards, while rehabilitation programmes (30–35% reduction) reduce long-duration and psychological claim costs. Coverage applies to injuries arising from mining operations, particularly machinery-related and overexertion incidents *(Appendix A.8)*, addressing both acute and cumulative risks, while exclusions include misconduct, unauthorised activities, criminal behaviour, and non-work-related or non-aggravated pre-existing conditions. Premiums are risk-adjusted by occupation and solar system and range from approximately Đ1.09M to Đ1.62M, ensuring pricing adequacy across risk profiles. Overall, the product balances cost control with adequate protection by reducing high-frequency claims while maintaining coverage for severe losses.

3 Summary of Pricing and Capital Modelling

Aggregate losses across all hazard areas were modelled using compound frequency-severity approaches, with claim frequency modelled using Poisson-based distributions and severity using lognormal-type distributions, reflecting differing risk profiles across hazard areas. Monte Carlo simulation (10,000 iterations) was used to generate aggregate loss distributions and estimate expected losses and tail risk measures. Stress testing was performed by applying shocks to frequency, severity and dependency to assess portfolio resilience under adverse conditions. Capital requirements were estimated using a Value-at-Risk (VaR) approach at the 99th percentile, capturing the impact of extreme but plausible loss scenarios.

3.1 Equipment Failure

Equipment failure exhibits a low-frequency, high severity risk profile. Before product implementation the mean annual aggregate loss was Đ229.2 M, with a 95th percentile of Đ237.8 M and a 99th percentile of Đ241.5 M. After applying the proposed product design *(Figure 1)*, the retained mean loss reduced to Đ128.2 M**,** while the 95th percentile fell to Đ132.2 M and the 99th percentile to Đ134.1 M, representing reductions of about 45% across the key measures. This indicates that the deductible and policy structure materially reduce both expected loss and tail risk.

Stress testing in *Table 1* shows that expected equipment failure losses rise materially as conditions worsen, but the proposed product continues to reduce retained losses across all scenarios. Retained loss falls from 55.91% of gross loss in the base case to 46.64% under moderate stress and 34.39% under severe stress, indicating that the product remains effective even as claim frequency and severity increase. This suggests the design provides meaningful protection not only under normal operating conditions but also in stressed environments. The full underlying expected loss values before and after product implementation are provided in *Appendix B.1*.

Portfolio premium is priced at Đ141.4 M, full derivation is provided in *Appendix B.2*, and premiums vary by solar system to reflect differences in underlying claim frequency and severity. These rates may be recalibrated over time as experience emerges, and risk conditions change.

3.2 Cargo Loss

The cargo loss portfolio exhibits a bimodal severity distribution where two distinct humps observed in (*Figure 2)* represent the occurrences of ordinary and catastrophic losses respectively. The aggregate cargo loss per exposure period is Đ 3.77 M with a standard deviation of Đ15.47 M. Moreover, the 95th percentile loss is approximately Đ 23.6 M, while the 99th percentile loss reaches approximately Đ 72.5 M. Since expected losses are moderate relative to shipment values, the distribution exhibits substantial tail risk driven by catastrophic cargo values.

By financial modelling ICPP, we can see from *Appendix B.3* that expected aggregate costs in the short term are approximately Đ 4.13 M, while expected returns are Đ4.84 M, resulting in expected net revenue of Đ 0.71 M. Over the long term, inflation increases both costs and returns, with expected costs rising to Đ 4.66 M and returns to Đ5.46 M, leading to slightly higher expected net revenue of Đ 0.80 M. A stable and consistent profit in both the short and long run demonstrates that ICPP is financially sustainable even under evolving economic conditions.

As seen in *Appendix B.4,* the implementation of ICPP will reduce expected losses by 32.77%, indicating partial risk transfer while still maintaining insured retention through deductibles. A 44% reduction in standard deviation equates to a significant decrease in loss volatility, which provides policyholders with greater financial stability. ICPP will also be extremely effective in reducing extreme losses, with the 99th percentile decreasing by 40%. This shows the proposed product is well designed to protect against catastrophic cargo loss events while preserving appropriate risk-sharing incentives.

The stress testing results demonstrate that the proposed insurance product provides increasing levels of protection as loss severity rises, as seen in *Table 1.* In stressed scenarios, the reduction increases to 42% and 53%, respectively, reflecting the product's stronger effectiveness in mitigating extreme losses. This highlights that the policy is well-designed to protect against catastrophic cargo loss events while preserving appropriate risk-sharing and pricing sustainability.

3.3 BI

The BI portfolio exhibits a low-frequency, high-severity risk profile with OCC preserving meaningful risk transfer for CQMC by limiting the insurer's exposure to smaller and less volatile losses. Through simulating different parameters the structure, the chosen was 21-day waiting period, Đ150,000 deductible, 80% coinsurance and Đ400,000 policy limit which materially reduces retained insurer liability.

Appendix B.5 revealed that before OCC was applied, the simulated portfolio revealed an expected aggregate loss of Đ8.50 B, standard deviation of Đ101.7 M and tail loss of Đ8.66 B at the 95th percentile and Đ8.73 B at the 99th percentile. Conversely, *Figure 3* reveals after OCC implementation, the aggregate loss evidently shifted downward with expected loss falling to Đ1.88 B showcasing substantial compression in both average losses and extreme outcomes. Thus, confirming OCC ability in achieving significant insurer-side risk reduction whilst retaining targeted risk transfer for severe interruption scenarios.

Table 1 reveals the stress testing for OCC remained effective under adverse conditions. Therefore, OCC improved affordability whilst reducing capital strain to deliver more efficient risk transfer through concentrating cover on high impact interruptions than routine events.

3.4 Workers' Compensation

The Workers' Compensation portfolio exhibits a relatively high-frequency, moderate - severity profile. The expected loss after design is Đ3.06M (with a standard deviation of Đ0.36M), which is a decrease from Đ3.51M before design. This reflects the impact of cost-

control features such as deductibles, waiting periods and benefit limits, which reduce claim payments and retained losses.

The distribution *(Figure 4)* is unimodal and slightly right-skewed with a peak around Đ3.50M, indicative of a stable and moderately dispersed loss profile. It has a 95th percentile of Đ3.68M and a 99th percentile of Đ3.95M *(Appendix B.6),* indicating moderate tail risk.

Stress testing in *Table 1* reveals that the SCPP design reduces retained losses across all scenarios. Expected loss falls from Đ5.26M to Đ4.59M under moderate stress and Đ8.93M to Đ7.76M under severe stress. This indicates that the design remains effective in reducing average insurer loss even under adverse conditions.

Capital requirements for SCPP are approximately Đ0.89M, compared with Đ0.36M pre-design. Although expected loss declines, capital increases because the remaining insured losses are more concentrated in larger claims, where tail risk is the primary driver of capital needs. Compared with other hazard areas, Workers' Compensation contributes less to extreme losses overall due to its lower severity and more predictable claim structure.

Premiums are risk-adjusted by occupation and solar system, ranging from approximately Đ1.09M to Đ1.62M across systems *(Appendix B.7).* While post-design premiums are lower, differences across solar systems remain, reflecting underlying variation in claim frequency and severity.

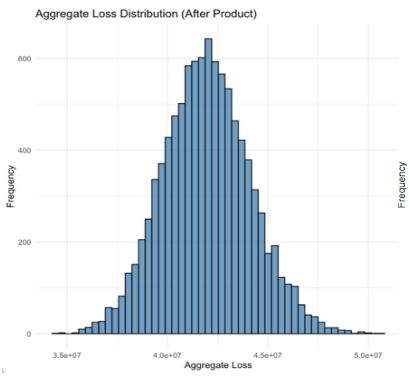


Figure 1: Post Equipment Loss Distribution

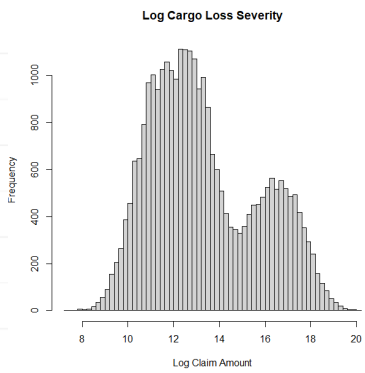


Figure 2: Post Log Cargo Loss Distribution

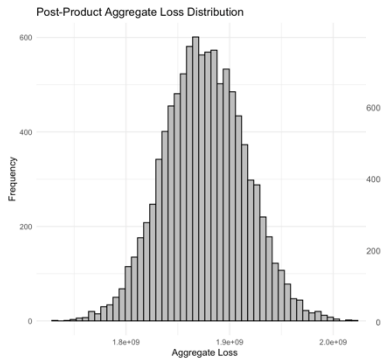


Figure 3: Post BI Loss Distribution

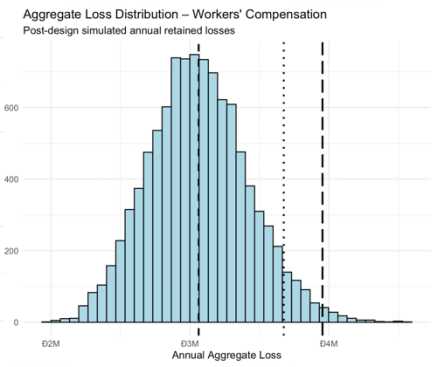


Figure 4: Post Workers' Comp Loss Distribution

Table 1: Percentage Reduction in Aggregate Loss

Scenario	Description	Equipment	Cargo	BI	Worker
Base	Attritional claims, no correlated events	55.91	32.77	84.66	12.76
Moderate stress (≈1-in-50)	+20% frequency, +25% severity	46.64	41.57	82.65	12.82
Severe stress (≈1-in-100)	+50% frequency, +75% severity	34.39	52.95	81.80	13.04

4 Risk Assessment

4.1 Risk Identification

Hazard Area	Zeta	Epsilon	Helionis Cluster

Workers Comp	**Pricing adequacy / Tail risk:** High workplace incident frequency leading to elevated claim counts and exposure to multi-injury events	**Reserving risk:** Moderate frequency with severity variability, creating uncertainty in claim outcomes	**Model risk:** Low observed claims with potential underestimation of emerging risks
Cargo Loss	**Severity / Tail risk:** Extreme radiation exposure cause damage to high-severity cargo, increasing tail risk and chances of catastrophic losses	**Operational risk:** Unstable debris regions increase navigation difficulty, leading to more micrometeoroid collisions and higher claim frequencies	**Correlation risk:** Dense asteroid clusters and erratic debris patterns lead to multi-shipment losses, leading to aggregated and severe loss exposure
BI	**Volatility risk:** Moderate frequency with material loss creating more volatile earnings especially during operational delays.	**Operational risk:** Highest claim frequency and loss average due to concentration risk as a large share of BI exposure is in this solar system.	**Operational Risk:** Lower observed interruption experience but with hidden vulnerabilities like supply chain dependency and resilience weakness.
Equipment	**Operational risk:** Loss experience is more unpredictable due to volcanic activity and unstable conditions, leading to greater variance in outcomes.	**Repair Complexity Risk:** Harsh conditions increase repair difficulty and downtime, leading to the highest loss severity and capital exposure among all systems.	**Operational Risk:** High failure frequency driven by dense debris fields and communication disruptions, increasing operational strain and maintenance uncertainty.

4.2 Correlated Risk Scenarios

Equipment failure losses may become correlated when a single event affects multiple assets or solar systems at once. Key scenarios include a radiation event that weakens shielding in Epsilon and Zeta, a shared equipment or software defect causing simultaneous breakdowns, and broader disruptions such as debris fragmentation, supply-chain shortages, or cyber-physical attack. These events matter because they create clustered failures and raise aggregate losses beyond independent claim expectations.

For cargo loss, large-scale solar radiation events can simultaneously impact multiple vessels, causing widespread cargo degradation and system failures. Additionally, major shifts in asteroid debris density can increase collision risk across multiple frequently used transport routes, leading to clustered loss events. Disruptions to shared communication and navigation systems can affect multiple transport vessels simultaneously, increasing probability of concurrent cargo losses.

For BI, a correlated scenario assumes that losses within the solar systems are caused from a shared external event rather than independently. BI reflects systemic disruption like solar storm, network-wide failures that would cause Helionis Cluster, Zeta and Epsilon outages and delays simultaneously. These disruptions could range from shared infrastructure or

software failure that interrupts multiple operations at once or broader interruptions like network breakdowns, supply chain and communication failure. When these situations occur, they create clustered downtime rather than isolated interruptions which can cause aggregate losses to surge about expected if claims occur independently. Thus, its crucial to include correlation in capital modelling, stress testing and pricing to decrease potential understated exposure.

For Workers' Compensation, losses may become correlated when a single event impacts multiple workers simultaneously, such as major machinery failures, hazardous exposures or systemic safety breakdowns across sites. These events lead to clustered claims, increasing aggregate losses beyond the independent expectations. Under a correlated shock, expected losses increased modestly, while tail risk rises significantly, indicating that correlation primarily drives extreme outcomes rather than average experience.

4.3 Scenario Testing

Under the best-case scenario, reduced frequency leads to lower losses across all hazards. In the moderate scenario, localised shocks (e.g. Zeta for Workers' Compensation and Equipment Failure and Epsilon for BI) result in moderate increases in expected losses, reflecting system-specific stress. In the worst-case scenario, where frequency and severity increase across all systems, losses rise significantly, particularly for Business Interruption and Equipment Failure due to their higher severity and tail risk. In contrast, Workers' Compensation shows smaller relative increases, reflecting its more stable and diversified loss profile.

	Scenario	Frequency Impact	Severity Impact	E[loss] Đ M
Equipment	Best case	$\lambda \times 0.75$	μ unchanged	76.06
	Moderate case (Zeta only)	$\lambda \times 1.15$ (Zeta only)	+20% severity (Zeta only)	126.25
	Worst case	$\lambda \times 2.0$	+50% severity	415.89

	Scenario	Frequency Impact	Severity Impact	E[loss] Đ M
Cargo Loss	Best case	$\lambda \times 0.75$	μ unchanged	2.79
	Moderate case (all systems)	$\lambda \times 1.15$ (all systems)	+20% severity (all systems)	5.57
	Worst case	$\lambda \times 2.0$ (all systems)	+50% severity	11.73

	Scenario	Frequency Impact	Severity Impact	E[loss] Đ M
BI	Best case	$\lambda \times 0.75$	μ unchanged	2275.21
	Moderate case (Epsilon only)	$\lambda \times 1.15$ (Epsilon only)	+20% severity (Epsilon only)	3502.78

Worst case	λ × 2.0	+50% severity	9100.43

	Scenario	Frequency Impact	Severity Impact	E[loss] Ð M
Worker Comp	Best case	λ × 0.75	μ unchanged	2.25
	Moderate case (Zeta only)	λ × 1.15 (Zeta only)	+20% severity (Zeta only)	3.48
	Worst case	λ × 2.0	+50% severity	8.98

4.4 Threat Table

Rank	Threat	Impact	Mitigation
1	Catastrophic workplace incidents (e.g. machinery failures, major accidents)	Drives tail risk (p99 losses) and potential multi-injury events, significantly increasing aggregate losses and capital requirements	Strengthen safety controls, preventive maintenance and planning/monitoring
2	Gravitational shear in asymmetric ring (Oryn Delta)	Causes sudden equipment instability and structural stress, increasing the likelihood of severe breakdowns, operational disruption, and large correlated losses	Apply a ring-zone rider with tighter coverage terms, higher deductibles, and stricter risk selection
3	High Asteroid Debris Density	Increases both claim frequency and severity by increasing probability of cargo losses and equipment failure.	Implement enhanced shielding requirements and stricter debris-monitoring protocols for operations in Helionis
4	High solar radiation exposure	Degrades cargo	Requires stronger radiation shielding, hardened storage systems and tighter operating limits
5	Supply Chain Interruption	Increases claim frequency	Diversify suppliers and maintain strong inventory buffers
6	Insufficient energy backup capacity	Reduces system resilience	Increase redundancy standards and backup capacity requirements

4.5 Risk Differences

The Helionis Cluster is generally exposed to higher operational strain, with dense debris, unstable conditions, and supply disruption increasing the likelihood of equipment damage,

cargo loss, worker injury, and BI. The presence of major asteroid clusters and erratic debris movement elevates collision risk for cargo vessels and mining infrastructure, contributing to frequent cargo losses, higher equipment breakdown risk, and more volatile operating conditions overall.

The Bayesia System is relatively more controlled, with mapped routes and more stable conditions reducing some physical and operational risks. However, high solar radiation remains a major hazard, particularly for high-value cargo and shielding-dependent equipment, while also posing worker safety concerns and the potential for interruption if systems are simultaneously affected.

Oryn Delta is the most fragile overall due to its combination of navigation instability, reduced visibility, and communication limitations. These factors increase the likelihood of misnavigation, delayed response to hazards, equipment failure, cargo damage, worker exposure to unsafe conditions, and longer operational interruptions.

Overall, Helionis Cluster and Oryn Delta present higher risk across the four hazards due to debris exposure and operational instability, while the Bayesia System is comparatively more resilient but still vulnerable to radiation-driven severity risk.

5 Ethical Considerations

1. Clear Outline of coverage

The clear communication of insurance terms (e.g. specified perils insured for, coverage and exclusions) avoid ambiguity for policyholders to allow for informed decision-making.

2. Fair Pricing

Loss premiums are determined using route-risk, ensuring that pricing reflects underlying hazards without unfair discrimination.

3. Environmental benefits

Deductible structures incentivise better maintenance practices and operational safety, indirectly reducing environmental damage and promoting sustainable operations.

4. Transparency of Assumptions

Key modelling assumptions (e.g. Poisson frequency, lognormal severity, inflation adjustments) should be clearly disclosed to ensure stakeholders understand the basis of pricing and risk estimates.

5. Affordability and Access

The product remained reasonably affordable so that the insurance is available to lower-risk or stronger operators. If the premiums are too high the firms most exposed to disruption would be excluded from protection and thus, raising concerns about equitable access to risk transfer.

6 Assumptions

a) Historical loss drivers remain relevant for future operations

Assumed to remain relevant for future interstellar mining operations, although actual risk levels may evolve as transportation networks expand.

b) Route risk classification reflects overall threat to cargo

It is assumed that the route risk rating (1 - 5) summarises multiple operational and environmental risks such as asteroid debris density, radiation exposure, and navigation complexity. Thus, the classification is used as an appropriate determinant for adjusting premiums in ICPP.

c) Expense ratio and risk loading

Expense ratio of 10%, risk margin of 15% and profit margin of 5% were assumed constant for all hazard areas for insurance pricing purposes. They reflect the underwriting expenses and compensation for capital and tail risk.

d) Stress Test and Scenario Test parameter adjustment choices

The best-case scenario is defined as lambda * 0.75 and mu unchanged.

e) Inflation is adequately captured by a 3-year moving average

Recent inflation is assumed to be a reasonable approximation for near-term claim cost escalation.

f) Historical claims are representative of near-term future experience

The historical equipment failure data are assumed to provide a reasonable basis for pricing CQMC's prospective business after adjusting for current inflation and product terms.

g) Solar system behaviour

No historical claims available for Bayesia and Oryn Delta hence, assumption that they behave broadly in line with existing historical systems.

7 Data and Data Limitations

a) Historical data are not CQMC's own direct claims history.

The claims data come from similar businesses written in other circumstances, so actual future experience may differ due to differences in workforce composition, operational practices and environmental conditions.

b) Limited system-specific data

Historical data are only available for Epsilon, Helionis Cluster and Zeta solar systems. For the other solar systems (Bayesia and Oryn Delta), risk characteristics are qualitative, introducing model uncertainty and potential misestimation of frequency and severity and, hence, product design.

c) Distribution modelling constraints

Claim frequency and severity are modelled using Poisson and lognormal distributions. Although these capture the overall shape of the data, extreme tail behaviour may not be fully represented.

d) Dependency and parameter uncertainty

Baseline modelling assumes independence across claims and systems, and key inputs (e.g., inflation, discount rates) are estimated from historical data. Changes in future conditions may increase aggregate losses and capital requirements.

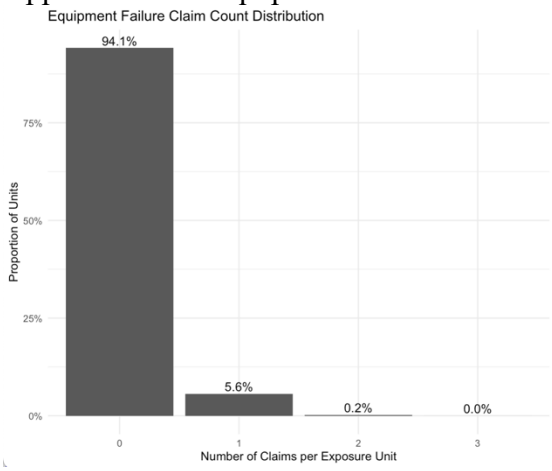
AI usage acknowledgement

AI tools were used to assist with drafting, code refinement, and wording suggestions. All modelling choices, calculations, assumptions, and interpretations were reviewed and validated against the provided datasets and case requirements by the team. AI outputs were not accepted without verification and were not relied upon as a substitute for actuarial judgement.

Appendix

Appendix A

Appendix A.1 - Equipment Failure Claim Count Distribution



Appendix A.2 - Equipment Severity Summary Table

Solar System	P25	Medium
Epsilon	44411.75	66246.5
Helionis Cluster	41550.00	60794.0
Zeta	49486.00	76119.0

Appendix A.3 - Equipment Tail Claim Table

Percentile 95	Percentile 99	Percentile 99.9	Max Claim
17953	275685.1	416083.8	761417.7

Appendix A.4 - Varying levels of premiums for the Interstellar Cargo Protection Policy depending on route risk.

Route Risk Level	Premium	Deductible %	Deductible
1	Đ 3,800,677	5%	190033.8
2	Đ 4,275,761	6.25%	267235.1
3	Đ 4,750,846	7.5%	356313.4
4	Đ 5225931	8.25%	457268.9
5	Đ 5701015	10%	570101.5

Appendix A.5 - Product Check Summary

Average Gross Claim	Average paid Claim	Payable proportion

765,918	146,105	0.565

Appendix A.6 - Comparison before and after OCC

Scenario	Expected Loss	P99	Required Premiums	ROC
Baseline	8,495,319,142	8,726,377,831	11,043,914,885	7.35
Post-Product	1,875,937,589	1,968,340,049	2,438,718,865	4.06

Appendix A.7 - Workers' Compensation Claim Duration and Severity Summary

Metric	Value
Number of claims	1,866
Mean claim duration (days)	56.8
Median claim duration (days)	7.0
75th percentile duration (days)	13.8
90th percentile duration (days)	55.0
95th percentile duration (days)	507.5
Maximum duration (days)	996
Mean claim amount (Đ)	7,826
Median claim amount (Đ)	2,020
75th percentile claim (Đ)	4,678
90th percentile claim (Đ)	15,467
95th percentile claim (Đ)	48,118
Maximum claim (Đ)	193,357

Appendix A.8 – Workers' Compensation Injury Mix

A.8A Injury Type Mix

Injury Type	Count	Proportion (%)
Cut laceration	655	35.1
Sprain, strain	574	30.8
Stress	211	11.3
Burns	174	9.3

Psychological	148	7.9
Other	95	5.1
Amputation	7	0.4

A.8B - Injury Cause Mix

Injury Cause	**Count**	**Proportion (%)**
Caught in machine	569	30.5
Overexertion	440	23.6
Vehicle accident	344	18.5
Stress/strain	297	15.9
Exposure	91	4.9
Other	81	4.4
Violence	41	2.2

Appendix B

Appendix B.1 - Equipment Stress-Testing Comparison

Scenario	**Expected loss before (Đ)**	**Expected loss after (Đ)**
Base case	229,168,937	101,036,161
1-in-50 year	343,826,551	183,471,522
1-in-100 year	601,712,258	394,761,500

Appendix B.2 – Premium Derivation & Premium by Solar System

Component	**Amount (Đ)**	**Basis**
Expected net loss	108,782,912	Simulation mean
Risk margin (15%)	16,317,437	Reflects parameter uncertainty and tail risk
Pure premium	**125,100,349**	**Sum of above**
Expense loading (10%)	10,878,291	Operational and administrative costs
Profit margin (5%)	5,439,146	Target return on premium
Gross premium	**141,417,786**	**Sum of above**
Per unit rate	6,171	Gross premium / 22,917 units

Solar system	Premium per exposure unit (Đ)	Portfolio premium (Đ)
Epsilon	5,446	50,353,188
Helionis Cluster	7,307	33,158,935
Zeta	6,391	58,364,239

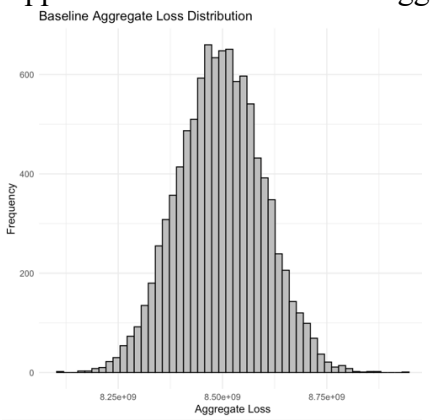
Appendix B.3 – Short-term (1 year) and long-term (5 years) costs, returns and net revenue for ICPP

Horizon	Costs	Returns	Net Revenue
Short-term	Đ 4,129,581	Đ 4,835,886	Đ 706,304.6
Long-term	Đ 4,664,347	Đ 5,462,115	Đ 797,768.4

Appendix B.4 – The mean, standard deviation, 95th percentile and 99th percentile of losses without and after implementing the ICPP product

Measure	Before Product (Gross Product)	After Product (Retained Loss)	Reduction Percent
Mean	Đ 3,926,116	Đ 2,639,642	32.77
Standard Deviation	Đ 15,475,467	Đ 8,718,374	43.66
95th Percentile	Đ 24,396,744	Đ 20,508,853	15.94
99th Percentile	Đ 77,838,253	Đ 46,192,879	40.66

Appendix B.5 – Pre-Product Aggregate Loss for BI



Appendix B.6 – Aggregate Loss Summary for Workers' Compensation (SCPP)

Scenario	Expected Loss (ĐM)	SD (ĐM)	p95 (ĐM)	p99 (ĐM)	Capital (ĐM)
Pre-design	3.51	0.15	3.77	3.87	0.36
Post-design	3.06	0.36	3.68	3.95	0.89

Appendix B.7 – Solar-system premium comparison for Workers' Compensation

Solar System	Expected Loss Pre (ĐM)	Expected Loss Post (ĐM)	Premium Pre (ĐM)	Premium Post (ĐM)	Premium Change (%)
Epsilon	1.15	0.91	1.50	1.18	-20.8
Helionis	0.86	0.84	1.12	1.09	-2.3
Zeta	1.49	1.25	1.93	1.62	-16.0

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